Transformée de Fourier

Exo 1:

1. Soit $f \in L^1(\mathbb{R})$, mq

$$\overline{\hat{f}}(\xi) = \hat{f}(-\xi) \quad , \xi \in \mathbb{R}$$

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2i\pi\xi x} f(x)\, dx$$

Soit $\xi \in \mathbb{R}$,

$$\overline{\hat{f}}(\xi) = \overline{\int_{\mathbb{R}} \underbrace{f(x)\, e^{-2i\pi\xi x}}_{f(x).\, e^{2i\pi\xi x}}\, dx}$$

$$\underset{(*)}{=} \int_{\mathbb{R}} f(x)\, e^{2i\pi\xi x}\, dx$$

(*) : rappel : • $f \in \mathcal{L}^1(X, \mathcal{B}, \mu)$, $f : X \to \mathbb{C}$

$$\int_X f \, d\mu = \int_X Re(f) \, d\mu + i \int_X Im(f) \, d\mu$$

bien définie car :

$$\int_X |Re(f)| \, d\mu \leq \int_X |f| \, d\mu < \infty$$

idem pour $Im(f)$

• $|\bar{f}(x)| = |f(x)|$,

$$\int_X \bar{f} \, d\mu = \int_X \overbrace{Re(\bar{f})}^{Re(f)} \, d\mu + i \int_X \overbrace{Im(\bar{f})}^{-Im(f)} \, d\mu$$

$$= \int_X Re(f) \, d\mu - i \int_X Im(f) \, d\mu$$

$$= \overline{\int_X f \, d\mu}$$

$$\Rightarrow \quad \hat{\bar{f}}(\xi) = \overline{\int_{\mathbb{R}} f(x)\, e^{-2i\pi(-\xi)x}\, dx}$$

$$= \overline{\hat{f}(-\xi)}$$

2. Supposons $f \in L^1(\mathbb{R})$ tq :

* f réelle, ie $(\forall x \in \mathbb{R})$: $f(x) \in \mathbb{R} \subset \mathbb{C}$

i.e. $(\text{p.p. } x \in \mathbb{R})$: $f(x) = \bar{f}(x)$

* f paire : $(\text{p.p. } x \in \mathbb{R})$: $f(-x) = f(x)$

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2i\pi \xi x} f(x)\, dx$$

$$= \underbrace{\int_{-\infty}^{0} e^{-2i\pi \xi x} f(x)\, dx}_{(1)} + \int_{0}^{\infty} e^{-2i\pi \xi x} f(x)\, dx$$

(1) $y = -x$

i.e. $x = -y$

$= \varphi(y)$, $\varphi : \mathbb{R}_+ \to \mathbb{R}_-$

$y \mapsto -y$

$$|\det \varphi'(y)| = |\varphi'(y)| = |-1| = 1$$

$$\hat{f}(\xi) = \int_{\mathbb{R}_+} e^{-2i\pi\xi(-y)} f(-y)\, |-1|\, dy$$

$$+ \int_{\mathbb{R}_+} e^{-2i\pi\xi x} f(x)\, dx$$

$$= \int_{\mathbb{R}_+} \left(e^{2i\pi\xi z} f(-z) + e^{-2i\pi\xi z} f(z) \right) dz$$

$$= \int_{\mathbb{R}_+} 2\cos(2\pi\xi z)\, f(z)\, dz$$] hypothèse f paire

Montrons en outre que $\hat{f}$ est réelle et paire :

$$* \quad \overline{\hat{f}(\xi)} = \overline{2 \int_{\mathbb{R}_+} \cos(2\pi \xi x)\, f(x)\, dx}$$

$$= 2 \int_{\mathbb{R}_+} \overline{\underbrace{\cos(2\pi \xi x)}_{\in \mathbb{R}} \cdot \underbrace{f(x)}_{\in \mathbb{R}}}\, dx$$

$$= \hat{f}(\xi) \in \mathbb{R}$$

Autre méthode :

1. $\Rightarrow \hat{\bar{f}}(\xi) = \overline{\hat{f}}(-\xi)$

$$\Rightarrow \overline{\hat{f}}(\xi) = \hat{\bar{f}}(-\xi)$$

$$= \hat{f}(-\xi) \quad \text{car } f = \bar{f}$$

$$= \hat{f}(\xi) \quad \text{si } \hat{f} \text{ paire (cf 2ème point)}$$

$$* \ \hat{f}(-\xi) = 2\int_{\mathbb{R}} \underbrace{\cos(-2\pi\xi x)}_{\cos(2\pi\xi x)} f(x)\,dx$$

$$= \hat{f}(\xi)$$

3. Supposons $f \in L^1(\mathbb{R})$ tq

(p.p. $x \in \mathbb{R}$) : $f(-x) = -f(x)$ imparité

et mq : $\hat{f}(\xi) = -2i \int_0^{\infty} \sin(2\pi\xi x) f(x)\,dx$

et que $\hat{f}$ est impaire ; mq si de plus

f réelle, $\hat{f}$ est imaginaire pure

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2i\pi\xi x} f(x)\,dx$$

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$y = -x$

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$= \varphi(y)$, $\varphi : \mathbb{R}_+ \to \mathbb{R}_-$

$y \mapsto -y$

$$|\det \varphi'(y)| = |\varphi'(y)| = |-1| = 1$$

$$\hat{f}(\xi) = \int_{\mathbb{R}_+} e^{-2i\pi\xi(-y)} f(-y)\,|-1|\,dy$$

$$+ \int_{\mathbb{R}_+} e^{-2i\pi\xi x} f(x)\,dx$$

$$= \int_{\mathbb{R}_+} \left(-e^{2i\pi\xi x} + e^{-2i\pi\xi x} \right) f(x)\,dx$$

$$= \int_{\mathbb{R}_+} -2i \sin(2\pi\xi x)\, f(x)\,dx$$

$$= -2i \int_{\mathbb{R}_+} \sin(2\pi \xi x) f(x)\, dx$$

$$\hat{f}(-\xi) = -2i \int_{\mathbb{R}_+} \underbrace{\sin(-2\pi \xi x)}_{-\sin(2\pi \xi x)} f(x)\, dx$$

$$= 2i \int_{\mathbb{R}_+} \sin(2\pi \xi x) f(x)\, dx$$

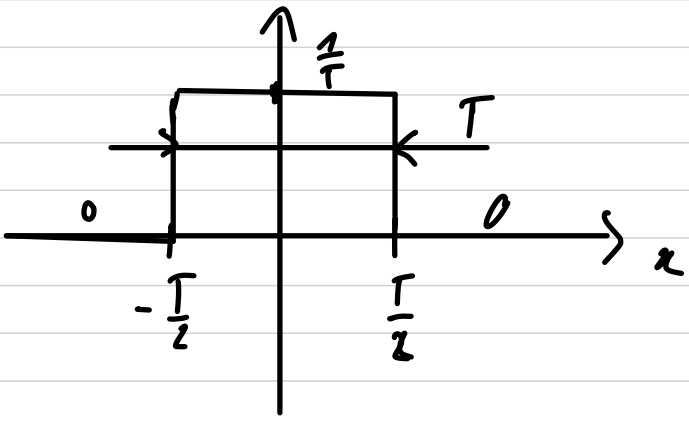
$$= -\hat{f}(\xi)$$

si f réelle, $f(x) \sin(2\pi \xi x) \in \mathbb{R}$

Donc $-2i \int_{\mathbb{R}_+} f(x) \sin(2\pi \xi x)\, dx \in i\mathbb{R}$

Exo 2 :

1. Soit $T > 0$,



$$f(x) = \frac{1}{T} \chi_{(-\frac{T}{2}, \frac{T}{2})}(x)$$

$$f \in L^1(\mathbb{R}) \quad \left(\text{cf} \int_{\mathbb{R}} \overset{=1}{|f|} \, dx < \infty \right)$$

Soit $\xi \in \mathbb{R}$,

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2i\pi \xi x} \cdot f(x) \, dx$$

$$= 2 \int_0^{\infty} \cos(2\pi \xi x)\, f(x)\, dx \Big) \quad \boxed{\text{paire en } \xi \text{ et réel}}$$

f paire

$$= 2 \int_0^{\frac{T}{2}} \cos(2\pi \xi x)\, \frac{1}{T}\, dx$$

$$= \left[\frac{2}{T} \cdot \frac{1}{2\pi \xi} \sin(2\pi \xi x) \right]_0^{\frac{T}{2}}$$

$\xi \neq 0$

$$= \frac{1}{T\pi \xi} \sin(T\pi \xi)$$

$$= \operatorname{sinc}(T\pi \xi) \qquad \left(\operatorname{sinc}(a) = \frac{\sin a}{a} \quad \sin 0 = 1 \right)$$

En $\xi = 0$, $\hat{f}(0) = 2 \int_0^{\frac{T}{2}} \frac{1}{T}\, dx = 1$

$$= \operatorname{sinc}(T\pi 0)$$

$$\Rightarrow \boxed{(\forall \xi \in \mathbb{R}) \quad . \ \hat{f}(\xi) = \operatorname{sinc}(\pi T \xi)}$$: paire et réelle

2. $f(x) = e^{-ax} \chi_{\mathbb{R}_+}$, $a > 0$

$$\left(\int_{\mathbb{R}} |f| \, dx = \int_0^{\infty} e^{-ax} \, dx = \left[-\frac{1}{a} \cdot e^{-ax} \right]_0^{\infty} = \frac{1}{a} < \infty \right)$$

soit $\xi \in \mathbb{R}$,

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2i\pi \xi x} f(x) \, dx$$

$$= \int_0^{\infty} e^{-2i\pi \xi x} e^{-ax} \, dx$$

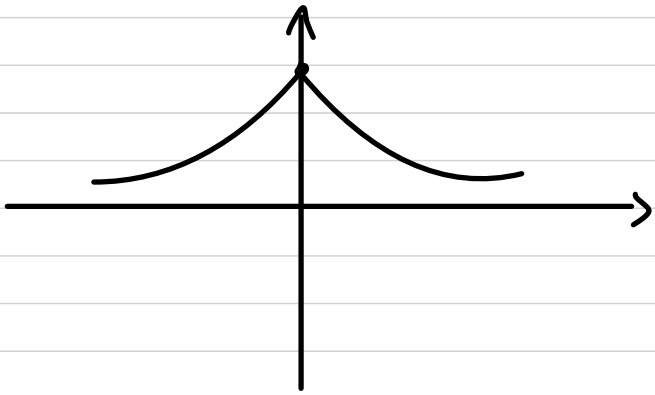
$$= \int_0^{\infty} e^{-x(2i\pi\xi + a)} \, dx$$

$$= \left[-\frac{1}{2i\pi\xi + a} e^{-x(2i\pi\xi + a)} \right]_0^{\infty}$$

$$= \left[- \frac{a - 2i\pi\xi}{a^2 + (2\pi\xi)^2} \, e^{-x(2i\pi\xi + a)} \right]_0^{\infty}$$

$$= \frac{a - 2i\pi\xi}{a^2 + 4\pi^2\xi^2}$$

3.



$f(x) = e^{-a|x|}$

$(a > 0)$

$f \in L^1(\mathbb{R})$ (idem 2.) ; f $\mathcal{C}^0(\mathbb{R})$

pas $\mathcal{C}^1(\mathbb{R})$

soit $\xi \in \mathbb{R}$

$$\hat{f}(\xi) = 2 \int_0^{\infty} \underbrace{\cos(2\pi\xi x)}_{\cos(-2\pi\xi x)} f(x) \, dx$$

doit être réel et pair

$$= 2 \int_0^{\infty} \mathrm{Re}\left(e^{-2i\pi\xi x} \right) e^{-ax} \, dx$$

$$= \mathrm{Re}\left(2 \int_0^{\infty} e^{-x(2i\pi\xi + a)} \, dx \right)$$

$$= \mathrm{Re}\left(2 \left[-\frac{1}{2i\pi\xi + a} e^{-x(2i\pi\xi + a)} \right]_0^{\infty} \right)$$

$$= \mathrm{Re}\left(2 \left[-\frac{a - 2i\pi\xi}{a^2 + (2\pi\xi)^2} e^{-x(2i\pi\xi + a)} \right]_0^{\infty} \right)$$

$$= 2 \, \mathrm{Re}\left(\frac{a - 2i\pi\xi}{a^2 + (2\pi\xi)^2} \right)$$

$$= \frac{2a}{a^2 + (2\pi\xi)^2}$$

$$\underset{|\xi| \to \infty}{\longrightarrow} 0 \left(\frac{1}{\xi^2} \right)$$

Exercice 3:

1. Soient $a, b \in \mathbb{R}$, $a \neq 0$

soit $f \in L^1(\mathbb{R})$, on pose $g(x) = f(ax+b)$

Mq $\hat{g}(\xi) = \dfrac{e^{2i\pi\xi\frac{b}{a}}}{|a|} \hat{f}\left(\dfrac{\xi}{a}\right)$

$f \in L^1(\mathbb{R}) \Rightarrow g \in L^1(\mathbb{R})$ et si $\xi \in \mathbb{R}$

$$\hat{g}(\xi) = \int_{\mathbb{R}} e^{-2i\pi\xi x} \overbrace{f(ax+b)}^{g(x)} \, dx$$

$y = ax+b$

i.e. $x = \varphi(y) = \dfrac{y-b}{a}$ $(a \neq 0)$

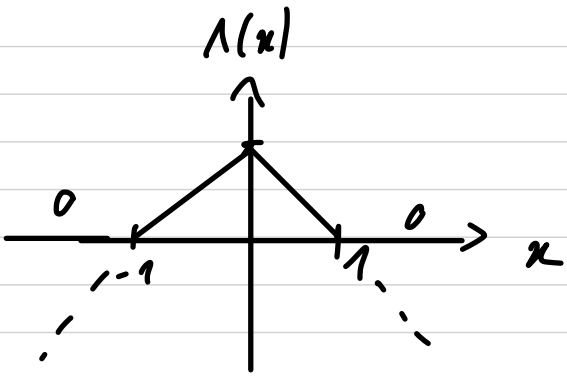
$|\det \varphi'(y)| = \dfrac{1}{|a|}$, $\varphi : \mathbb{R} \to \mathbb{R}$

$$\Rightarrow \hat{g}(\xi) = \int_{\mathbb{R}} e^{-2i\pi\xi\left(\frac{y-b}{a}\right)} f(y) \frac{1}{|a|} \, dy$$

$$= \frac{e^{2i\pi\xi\frac{b}{a}}}{|a|} \hat{f}\left(\frac{\xi}{a}\right)$$

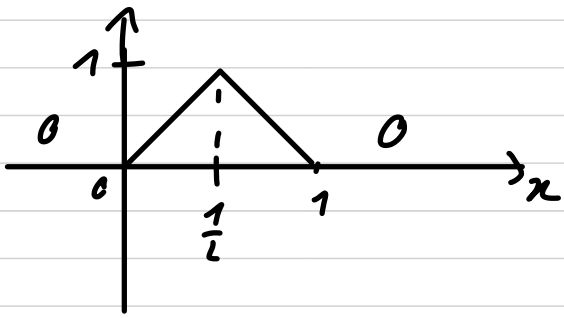
2. $\Lambda(x) = (1-|x|)\, \chi_{[-1,1]}(x)$

Λ $\mathcal{C}^0$ pas $\mathcal{C}^1$



$\widehat{\Lambda(2x-1)}$

$2\left(x-\frac{1}{2}\right)$



On utilise la parité de Λ et le 1 :

$$\hat{\Lambda}(\xi) = 2\int_0^{\infty} \cos(2\pi \xi x)\, \Lambda(x)\, dx$$

$$= 2\int_0^1 \underbrace{\cos(2\pi \xi x)}_{v'} \underbrace{(1-x)}_{u}\, dx$$

$$\underset{\xi \neq 0}{=} 2\left[\frac{\sin(2\pi\xi x)}{2\pi\xi}(1-x)\right]_0^1 - 2\int_0^1 \frac{\sin(2\pi\xi x)}{2\pi\xi}(-1)\,dx$$

$$= 2\left[-\frac{\cos(2\pi\xi x)}{(2\pi\xi)^2}\right]_0^1$$

$$= \frac{1}{2\pi^2\xi^2}\Big(1 - \underbrace{\cos(2\pi\xi)}_{1-2\sin^2(\pi\xi)}\Big)$$

$$= \frac{\sin^2(\pi\xi)}{\pi^2\xi^2}$$

$$= \operatorname{sinc}^2(\pi\xi) \quad \leftarrow \xi = 0 \text{ inclus}$$